Machine Learning for Real Estate: House Price Estimation Analysis

Project Report

1. Introduction

1.2 Project Objectives

- To clean and preprocess raw real estate data, handling significant missing values in columns like 'Takas' and 'Eşya_Durumu' etc.
- To visualize price-area relationships and identify market trends.
- To implement and tune advanced regression models, specifically LightGBM, to minimize prediction error (MAE/RMSE).

2. Data Preprocessing and Cleaning

2.1 Data Cleaning Strategy

The initial dataset contained 20,326 entries. The cleaning process was as follows:

1. **Row Removal:** Rows with 4 or more null values were deleted to maintain data integrity.
2. **Imputation Logic:** Missing 'Oda_Sayısı' (Room Count) values were filled using the logic: Room Count = Bathroom Count + 1.
3. **Categorical Handling:** Missing values in 'Takas' were set to 'Yok' and the rest of the null values were set to 'Bilinmiyor' (Unknown) to prevent overfitting.
4. **Outlier Removal:** Properties with unrealistic 'Net_Metrekare' (<10 m^2 or >2000 m^2) were removed to prevent skewing the model unrealistic room count, bathroom count, prices etc.

2.2 Feature Engineering

categorical variables such as 'Bulunduğu_Kat', 'Isıtma_Tipi', and 'Şehir' were transformed using **One-Hot Encoding** (pd.get_dummies), resulting in a final feature set of 124 columns.

3. Methodology

3.1 Model Selection

Several regression algorithms were considered, including Lasso, Decision Trees, SVM, and Random Forest. **LightGBM (Light Gradient Boosting Machine) ** was selected as the primary model due to its significantly better results.

3.2 Hyperparameter Tuning

To optimize the LightGBM model, **RandomizedSearchCV** was employed. Grid and Bayesian optimization was also tested but RandomCV was found to have the best accuracy/result. The search spanned 30 iterations with 3-fold cross-validation. The best parameters found were:

- Learning Rate: 0.05
  Num Leaves: 15
  N Estimators: 200
  Subsample: 0.8

4. Results and Evaluation

4.1 Performance Metrics

The model was evaluated on a test set (20% split) using the following metrics:

Metric	Value
R^2 Score (Accuracy)	70.19%
MAE (Mean Absolute Error)	685,350 TL
RMSE (Root Mean Squared Error)	1,369,304 TL
Training Score	77.66%

4.2 Analysis

The model achieved an R^2 score of 70.19% on the test data, which is close to the training score of 77.66%, indicating a well-balanced model with minimal overfitting. The MAE of roughly 685k TL suggests that, on average, the model's predictions are within this range of the actual sale price. Given the high variance in property prices (max price vs. min price), this represents a solid baseline for estimation.

5. Conclusion

By effectively handling missing data and utilizing LightGBM with Randomized Search, we achieved a semi-good accuracy result. Future improvements could involve integrating location-based geospatial data or more granular neighborhood statistics to further reduce the RMSE and increase accuracy.